

CAN-Based Engine Monitoring & Vehicle Control System

Distributed Multi-Node Automotive Network with Dynamic Fault Detection, LPC2129 ARM7 Microcontrollers, and CAN 2.0B Protocol

Architecture:	3-Node Distributed CAN Bus (250 kbps)
Fault Detection:	Software Watchdog with Node Error Screens
Microcontroller:	NXP LPC2129 (ARM7TDMI-S @ 60MHz)
Display Interface:	20×4 Alphanumeric LCD (HD44780 on P0.0-P0.10)
Sensors & Actuators:	DS18B20 (P0.22), GP2D12 (P0.27), 8-LEDs (P0.0-P0.7)

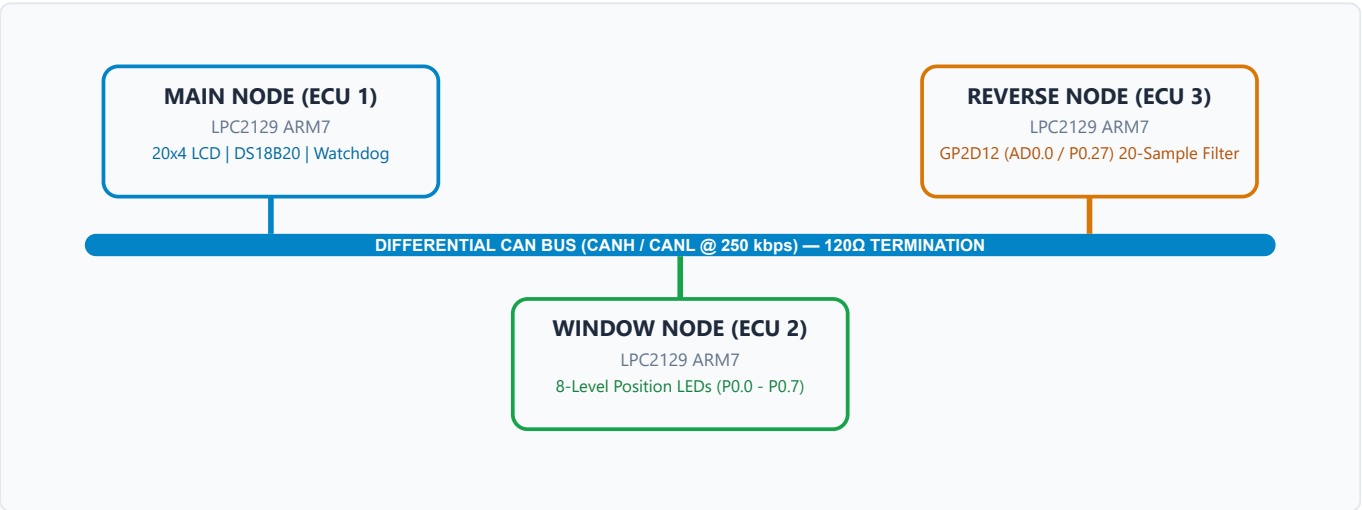
1. System Overview & Architecture

This project implements an intelligent distributed automotive Electronic Control Unit (ECU) network using three **NXP LPC2129 ARM7** microcontrollers communicating over an ISO 11898 compliant **CAN 2.0B bus at 250 kbps** with active node failure detection and automatic dashboard recovery.

1.1 Distributed ECU Nodes & Fault Supervision

- **Main Telemetry Node (Master Instrument Cluster):** Manages the 20×4 LCD digital instrument cluster, samples engine temperature from a DS18B20 1-Wire sensor, supervises CAN response timeouts, and displays dedicated error screens upon node disconnection.
- **Window Glass Control Node (Body Control ECU):** Features 8 physical LED levels (P0.0-P0.7) supporting single-tap step increments, continuous press holding, and physical boundary protections.
- **Reverse Alert Node (ADAS / Parking Proximity ECU):** Employs a 20-sample trimmed mean filter, digital low-pass filtering, and deadband hysteresis on the Sharp GP2D12 IR sensor (AD0.0 / P0.27).

1.2 System Architecture Block Diagram



2. CAN Communication Protocol Matrix

CAN Identifier	Frame Type	DLC	Source ECU	Destination	Payload Definition	Timeout & Error Handling
0x101	Data Frame	1 Byte	Main Node	Window Node	0x01=UP / 0x02=DOWN	500ms response timeout
0x102	Data Frame	1 Byte	Main Node	Reverse Node	0x01=ENABLE / 0x00=DISABLE	Activates/Deactivates reverse proximity sensing
0x103	Data Frame	2 Bytes	Reverse Node	Main Node	Byte 0: Filtered Distance (cm) Byte 1: Zone ID (1..4)	800ms watchdog with automatic recovery

CAN Identifier	Frame Type	DLC	Source ECU	Destination	Payload Definition	Timeout & Error Handling
0x201	Data Frame	2 Bytes	Window Node	Main Node	Byte 0: Level (0..8) Byte 1: Percentage (0..100%)	Acknowledges exact window level position

3. 20×4 LCD Visual Screens Gallery & Error Displays

1. SYSTEM BOOT / INITIALIZATION SCREEN

```
=====
SMART VEHICLE ECU
ENGINE & SAFETY OS
=====
```

2. DASHBOARD - NORMAL & ALL ECUS OK

```
--- VEHICLE DASH ---
TEMP  : 32.5°C [ OK ]
STATUS: ALL ECUS OK
WIN: 50%   REV: OFF
```

3. DASHBOARD - ECU OFFLINE DETECTED

```
--- VEHICLE DASH ---
TEMP  : 32.5°C [ OK ]
! WARN: ECU OFFLINE
WIN: 50%   REV: OFF
```

4. DASHBOARD - TEMP SENSOR FAULT

```
--- VEHICLE DASH ---
TEMP  : SENSOR ERR
STATUS: CHECK 1-WIRE
WIN: 50%   REV: OFF
```

5. WINDOW MODULE DISCONNECTED

```
! CAN BUS WARNING !
NODE: WINDOW MODULE
ERROR: NO RESPONSE
STATUS: OFFLINE /ERR
```

6. REVERSE RADAR DISCONNECTED

```
! CAN BUS WARNING !
NODE: REVERSE RADAR
ERROR: NO RESPONSE
STATUS: OFFLINE /ERR
```

7. POWER WINDOW - LEVEL 4 (50% GLASS)

```
=== POWER WINDOW ===
MOTION: ROLLING UP ▲
GLASS: [||||...]50%
STATUS: LEVEL 4 OF 8
```

8. REVERSE RADAR - SAFE ZONE (≥80 cm)

```
<< REVERSE RADAR >>
DIST  : 110 cm[SAFE]
ZONE  : GREEN / OK
BAR   :[|||||||||]
```

4. Complete Hardware Pin Connections (Exact Hardware Mapping)

4.1 Main Node (Master ECU) Pin Mapping

LPC2129 Port / Pin	Connected Peripheral	Peripheral Pin / Function	Description & Circuit Requirements
P0.0 – P0.7	20×4 LCD Display	D0 – D7	8-Bit Parallel Data Bus (Data mask: 0xFF)
P0.8	20×4 LCD Display	EN (Enable)	Active High Strobe Pulse
P0.9	20×4 LCD Display	RS (Register Select)	0 = Command Mode, 1 = Data Mode
P0.10	20×4 LCD Display	RW (Read / Write)	0 = Write Operation (Tied to GND / Low)
P0.22	DS18B20 Temp Sensor	DQ (Data Line)	1-Wire Data Line with 4.7kΩ Pull-Up to +5V
P0.16 (EINT0)	Window UP Pushbutton	Terminal 1	Active LOW (Single tap increment / Continuous hold)
P0.14 (EINT1)	Window DOWN Pushbutton	Terminal 1	Active LOW (Single tap decrement / Continuous hold)
P0.15 (EINT2)	Reverse Gear Switch	Terminal 1	Active LOW Reverse Engage Switch
P0.24 (TD1)	CAN Transceiver (MCP2551)	TXD (Pin 1)	CAN Controller 1 Transmit Line
P0.25 (RD1)	CAN Transceiver (MCP2551)	RXD (Pin 4)	CAN Controller 1 Receive Line

4.2 Window Glass Control Node Pin Mapping

LPC2129 Port / Pin	Connected Peripheral	Peripheral Function	Description
P0.0 – P0.7	Window Position Actuator / LED Bar	Position Indicators (LED 0..7)	8-Level discrete position indicator bar (Active LOW)
P0.24 / P0.25	CAN Transceiver (MCP2551)	TD1 / RD1	CAN Bus Interface (250 kbps)

4.3 Reverse Alert Node Pin Mapping

LPC2129 Port / Pin	Connected Peripheral	Peripheral Function	Description
P0.27 (AD0.0 / AIN0)	Sharp GP2D12 IR Sensor	Vo (Analog Out)	10-bit ADC with 20-sample trimmed mean filter & hysteresis
P0.24 / P0.25	CAN Transceiver (MCP2551)	TD1 / RD1	CAN Bus Interface (250 kbps)

5. Build & Proteus Simulation Instructions

1. **Keil uVision:** Open each node project, verify target device is **NXP LPC2129**, compile with crystal frequency at 12.0 MHz, and generate .hex files.
2. **Proteus Simulation:**
 - Place three **LPC2129** microcontrollers and three **MCP2551** transceivers.
 - Connect differential pair CANH and CANL across all three nodes with two **120Ω termination resistors**.
 - Wire the **20×4 LCD to P0.0–P0.10**, **DS18B20 to P0.22**, **LED bar to P0.0–P0.7**, and **GP2D12 to P0.27**.
 - Load the compiled .hex files and test disconnecting any CAN line to verify live node disconnection error alerts.